2) a) $\alpha f_p(x) = h_p(x)$

$$h_p = \alpha(f_p(x)) = \alpha(f_p(x+T)) = \alpha f_p(x+T) = h_p(x+T)$$

b) $\alpha + f_p(x) = h_p(x)$

$$h_p(x) = \alpha + f_p(x) = \alpha + f_p(x+T) = h_p(x+T)$$

$\vdots$

e) $f_p(x) \cdot g_p(x) = h_p(x)$

$$\Rightarrow f_p(x) \cdot g_p(x) = f_p(x+T) \cdot g_p(x+T) = h_p(x+T) = h_p(x)$$

4) $$L \cdot \sum_{n=-\infty}^{\infty} \delta[x - nL] = \sum_{n=-\infty}^{\infty} e^{in \frac{2\pi}{L} \cdot x}$$

$f(x) = \sum_{n=-\infty}^{\infty} \delta[x - nL]$, periodic function with $T = L$

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in \frac{2\pi}{L} x}$$

$$c_n = \frac{1}{L} \int_{-\frac{L}{2}}^{\frac{L}{2}} f(x) e^{-in \frac{2\pi}{L} x} dx = \frac{1}{L} \int_{-\frac{L}{2}}^{\frac{L}{2}} \delta(x) \cdot e^{-in \frac{2\pi}{L} x} =$$

$$= \frac{1}{L} \cdot e^{-in \frac{2\pi}{L} \cdot 0} = \frac{1}{L} \cdot 1 = \frac{1}{L}$$

$$\Rightarrow f(x) = \sum_{n=-\infty}^{\infty} \frac{1}{L} \cdot e^{in \frac{2\pi}{L} x} = \frac{1}{L} \sum_{n=-\infty}^{\infty} e^{in \frac{2\pi}{L} \cdot x} \Rightarrow$$

$$\Rightarrow L \cdot f(x) = \sum_{n=-\infty}^{\infty} e^{in \frac{2\pi}{L} x}$$ true

5) a) $\mathcal{F}[f(t)] = \hat{f}(t) = \int_{-\infty}^{\infty} f(t)\, e^{-i\omega t}\, dt$

$$\mathcal{F}[f(at)] = \int_{-\infty}^{\infty} f(at) \cdot e^{-i\omega t}\, dt$$

$$u = at \Rightarrow t = \frac{u}{a}$$

$$dt = \frac{du}{a}$$

$$\Rightarrow \int f(u) \cdot e^{-i\omega \frac{u}{a}} \frac{du}{a} =$$

case $a > 0 \Rightarrow \frac{1}{a}\int_{-\infty}^{\infty} f(u)\, e^{-i\frac{\omega}{a}u}\, du = \frac{1}{a}\hat{f}\left(\frac{\omega}{a}\right)$

case $a < 0 \Rightarrow \int_{\infty}^{-\infty} f(u)\, e^{-i\frac{\omega}{a}u} \frac{du}{a} = -\frac{1}{a}\int_{-\infty}^{\infty} f(u)^{-i\frac{\omega}{a}u}\, du = \frac{1}{|a|}\hat{f}\left(\frac{\omega}{a}\right)$

$\Rightarrow \frac{1}{|a|}\hat{f}\left(\frac{\omega}{a}\right)$ true

6) a) $x(t)$ real $\Rightarrow \hat{x}(f) = \int_{-\infty}^{\infty} x(t)\, e^{-i2\pi f t}\, dt$

$$\hat{x}^*(f) = \int_{-\infty}^{\infty} x^*(t) \left(e^{-i2\pi f t}\right)^* dt = \int_{-\infty}^{\infty} x(t)\, e^{i2\pi f t}\, dt$$

$$\hat{x}(-f) = \int_{-\infty}^{\infty} x(t)^{-i2\pi(-f)t}\, dt = \int_{-\infty}^{\infty} x(t)\, e^{i2\pi f t}\, dt$$

b) $x(t)$ imaginary $\Rightarrow \hat{x}(f) = \int_{-\infty}^{\infty} x(t)\, e^{-i2\pi f t}\, dt$

$$\hat{x}^*(f) = \int_{-\infty}^{\infty} x^*(t) \left(e^{-2\pi f t}\right)^* dt = \int_{-\infty}^{\infty} -x(t)\, e^{i2\pi f t}\, dt$$

$$\hat{x}(-f) = \int_{-\infty}^{\infty} x(t)\, e^{-i2\pi(-f)t}\, dt = \int_{-\infty}^{\infty} x(t)\, e^{i2\pi f t}\, dt$$

c) $x(t) = x(-t) \Rightarrow \hat{x}(f) = \int_{-\infty}^{\infty} x(t)\, e^{-i2\pi f t}\, dt$

$$\left.\hat{x}(-f) = \int_{-\infty}^{\infty} x(t)\, e^{i2\pi f t}\, dt \right\} \Rightarrow$$

$$\tau = -t \Rightarrow dt = -d\tau$$

$$\Rightarrow \int_{\infty}^{-\infty} x(-\tau)\, e^{i2\pi f(-\tau)}(-d\tau) = \int_{-\infty}^{\infty} x(-\tau)\, e^{-i2\pi f \tau}\, d\tau =$$

$$= \int_{-\infty}^{\infty} x(\tau)\, e^{-i2\pi f \tau}\, d\tau$$

d) $x(-t) = -x(t) \Rightarrow \hat{x}(f) = \int_{-\infty}^{\infty} x(t)\, e^{-i2\pi f t}\, dt$

$$\hat{x}(-f) = \int_{-\infty}^{\infty} x(t)\, e^{-i2\pi(-f)t}\, dt =$$

$$t = -\tau \Rightarrow dt = -d\tau$$

$$= \int_{\infty}^{-\infty} x(-\tau)\, e^{i2\pi f(-\tau)}(-d\tau) =$$

$$= \int_{-\infty}^{\infty} -x(\tau)\, e^{-i2\pi f \tau}\, d\tau$$

7) a) $f * g = \int_{-\infty}^{\infty} f(\tau)\, g(t-\tau)\, d\tau = \int_{-\infty}^{\infty} f(t-u)\, g(u)\, du = g * f$

$$u = t - \tau$$

$$\tau = t - u$$

$$du = dt$$

b) $\mathcal{F}[f*g] = \int_{-\infty}^{\infty} (f(t) * g(t))\, e^{-iwt}\, dt = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\tau)\, g(t-\tau)\, d\tau\, e^{-iwt}\, dt =$

$$= \int_{-\infty}^{\infty} f(\tau) \int_{-\infty}^{\infty} g(t-\tau)\, e^{-iwt}\, dt\, d\tau = \int_{-\infty}^{\infty} f(\tau) \int_{-\infty}^{\infty} g(u)\, e^{-iw(u+\tau)}\, du =$$

$$u = t - \tau \Rightarrow t = u + \tau$$

$$du = dt$$

$$= \int_{-\infty}^{\infty} f(\tau) \left(\int_{-\infty}^{\infty} g(u) \cdot e^{-iwu} \cdot e^{-iw\tau}\, du \right) d\tau = \int_{-\infty}^{\infty} f(\tau)\, e^{-iw\tau} \left(\int_{-\infty}^{\infty} g(u)\, e^{-iwu}\, du \right) d\tau =$$

$$= \int_{-\infty}^{\infty} f(\tau) \cdot e^{-iw\tau} \cdot g(w)\, d\tau = g(w) \cdot \int_{-\infty}^{\infty} f(\tau) \cdot e^{-iw\tau}\, d\tau =$$

$$= g(w) \cdot f(w) = f(w) \cdot g(w)$$

d) $f * \delta = \int_{-\infty}^{\infty} f(\tau) \cdot \delta(t-\tau)\, d\tau = f(t)$

e) $\left. \begin{array}{l} f * \delta(t-t_0) = \int_{-\infty}^{\infty} f(\tau)\, \delta(t-t_0-\tau)\, d\tau \\ t - t_0 - \tau = 0 \Rightarrow \tau = t - t_0 \end{array} \right\} \Rightarrow f(t-t_0)$